

16-867

Human Robot Interaction

Introduction

Instructor: Andrea Bajcsy

Welcome!

Professor



Andrea Bajcsy
(BYE-chee)

What to call me:

- Andrea (*if you are a grad student*)
- Prof. Bajcsy or Prof. B (*if you are undergrad*)

Office Location: NSH 4629

Office Hours: Tuesdays, 12:20-1:20pm (*after class*)

Email: abajcsy@cmu.edu

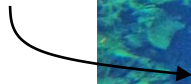
Professor



Andrea Bajcsy

Fun fact: I swam between the Eurasian and North American tectonic plates!

North American



Eurasian



Silfra Fissure, Iceland

Teaching Assistants



Hyun Joe Jeong, PhD Student

Research Interests:

- Learning task-relevant representations for robot decision-making
- Generative modeling, representation learning, and RL

Office Location: NSH **TBA**

Office Hours: **TBA**

- **please take survey on Canvas so we can select OHs that suit folks best!**

Email: hyunjoej@andrew.cmu.edu

Teaching Assistants



Hyun Joe Jeong, PhD Student

Fun fact: "I love baking!"



Joe's flan!

Teaching Assistants



Sreyas Venkataraman,
MSR Student

Research Interests:

- Sample-efficient learning and adaptation of foundation models
- Data curation, representation learning, generative modeling, architectural design, RL.

Office Location: NSH **TBA**

Office Hours: **TBA**

- **please take survey on Canvas so we can select OHs that suit folks best!**

Email: sreyasv@andrew.cmu.edu

Teaching Assistants



Sreyas Venkataraman,
MSR Student

Fun fact: “I love playing tennis (very much at the amateur level)”



What is next?

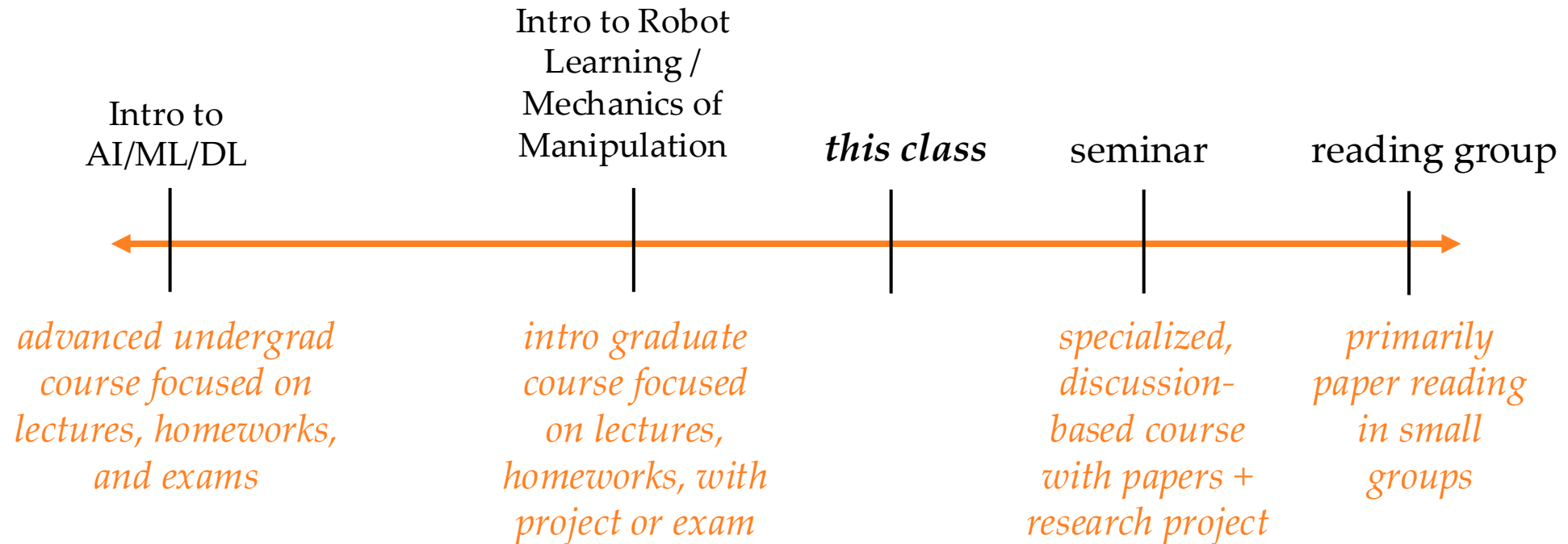
Course Content

Logistics

Intro Survey

(Intro to Single-Agent Decision Making)

What exactly is this class?



What makes human-robot interaction different from “typical” robotics?



Small group activity (5 min)

Turn to your neighbor(s), introduce yourself, and discuss:

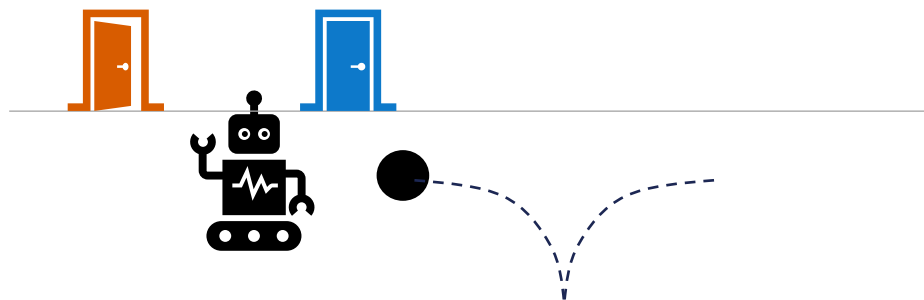
What makes human-robot interaction different from “typical” robotics?



vs.

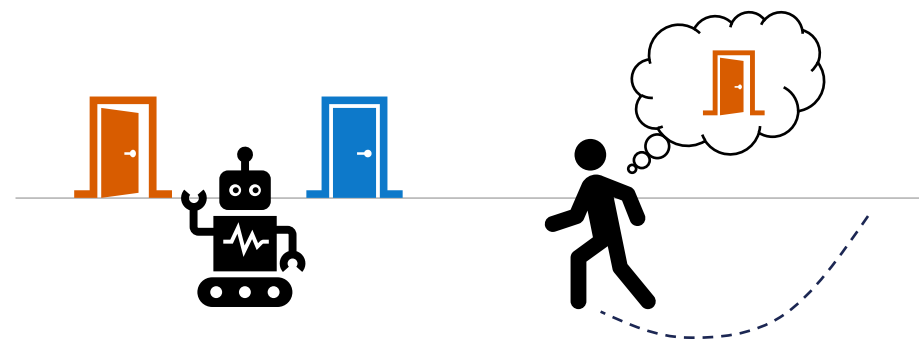


Interaction



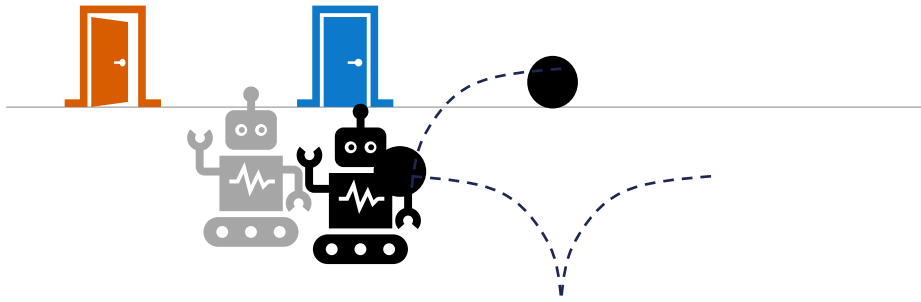
Environment driven by
laws of physics

Human Interaction



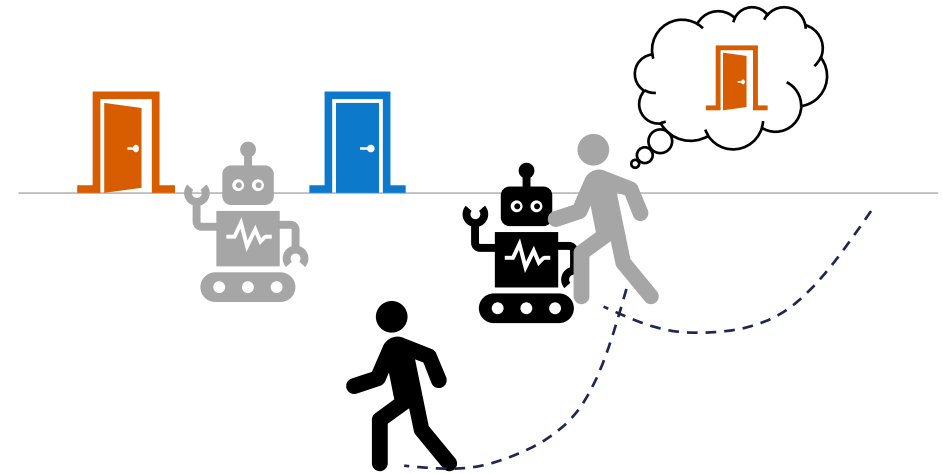
Human driven by physics and hidden
internal objectives

Interaction



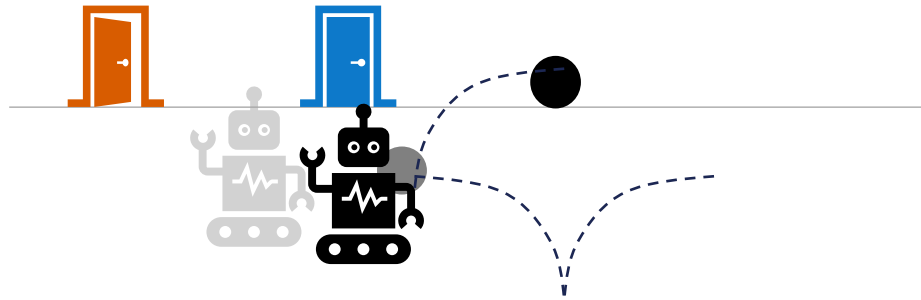
Environment can *be influenced* by
robot's actions **directly**

Human Interaction



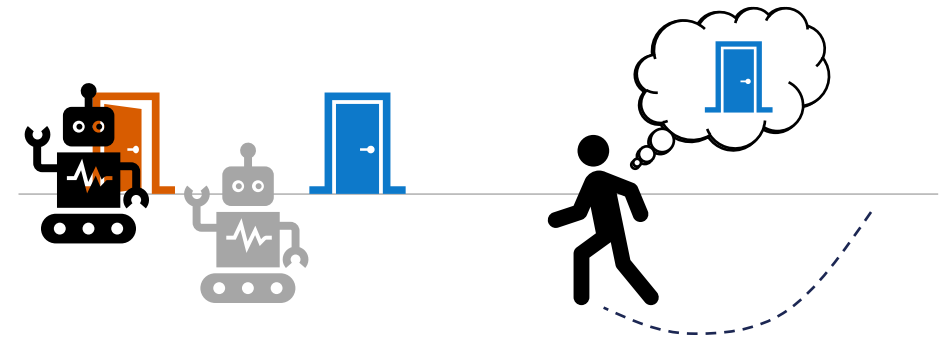
Human can *be influenced* by the robot's
actions **directly....**

Interaction



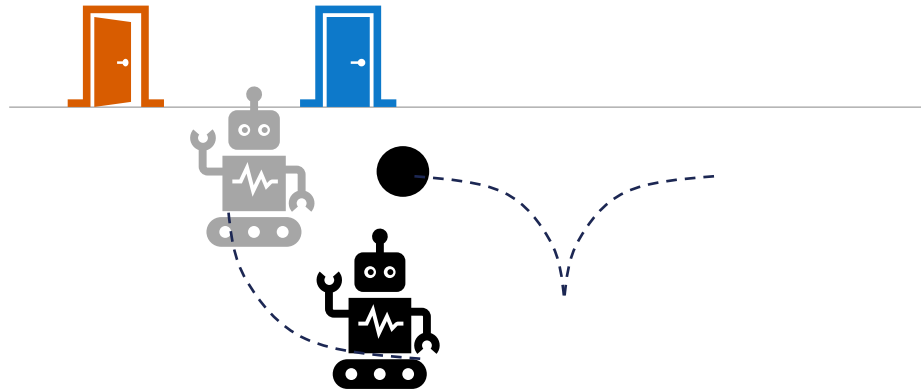
Environment can *be influenced* by
robot's actions **directly**

Human Interaction



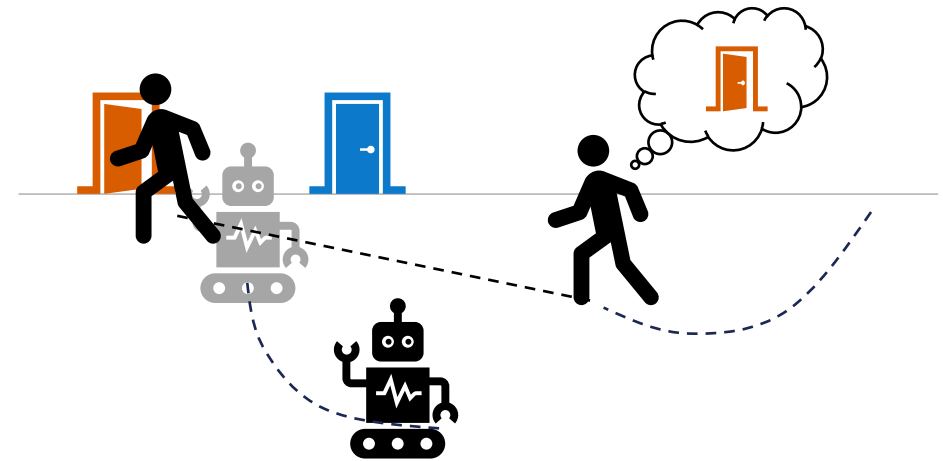
Human can *be influenced* by the robot's
actions and **indirectly**

Interaction



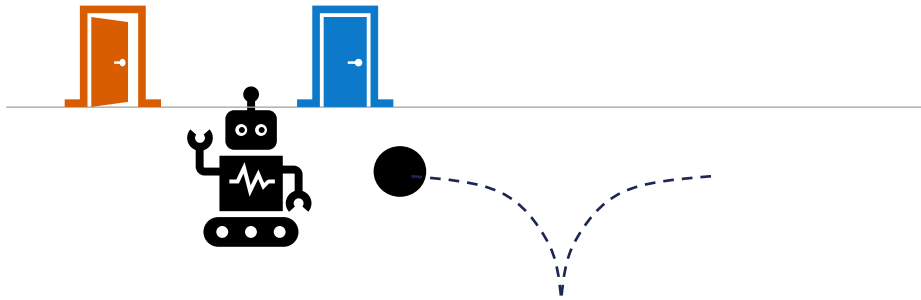
Environment can *influence* the robot's actions **indirectly**

Human Interaction



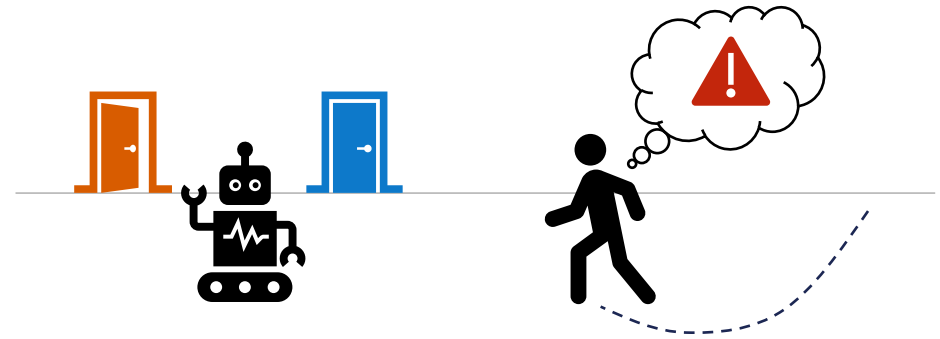
Human can *influence* the robot's behavior **directly or indirectly**

Interaction



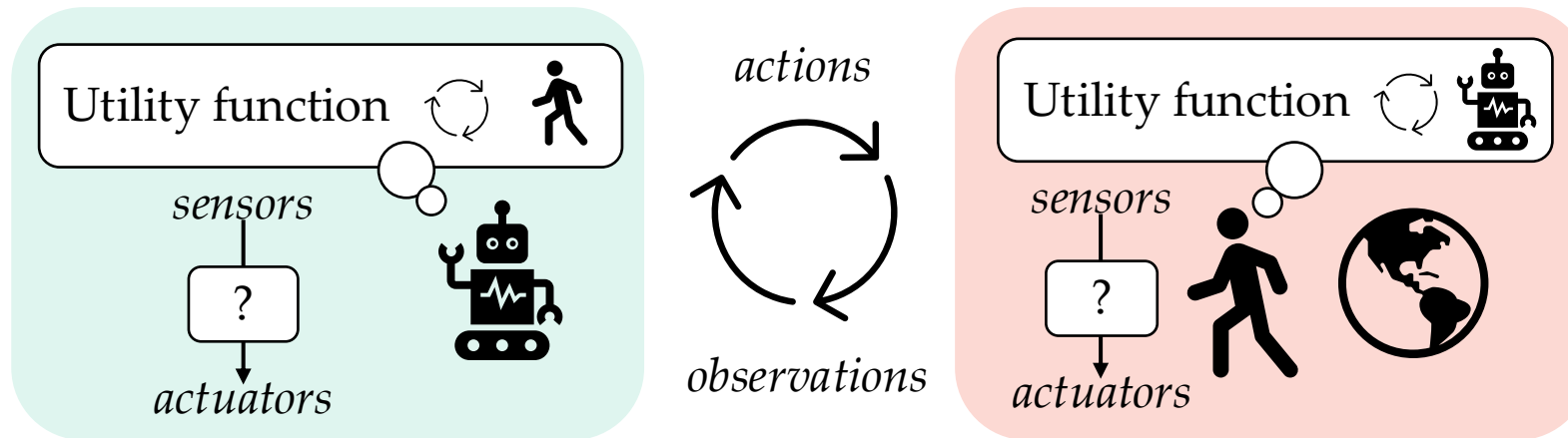
“Environment” is not a stakeholder

Human Interaction



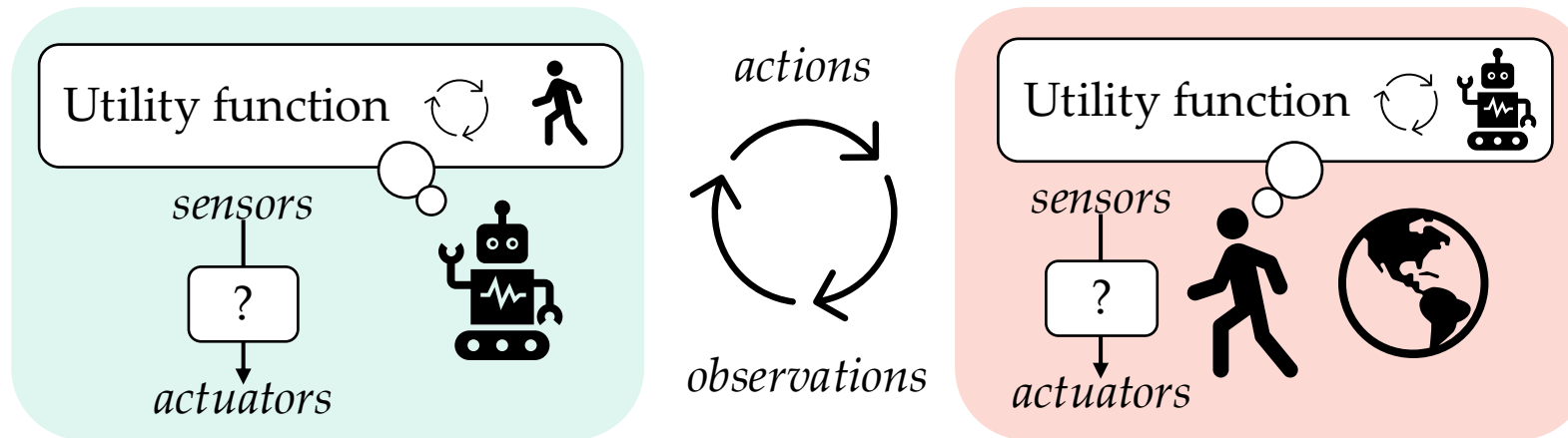
Human is a stakeholder! (e.g., wants to derive value from robot)

How is algorithmic HRI different?



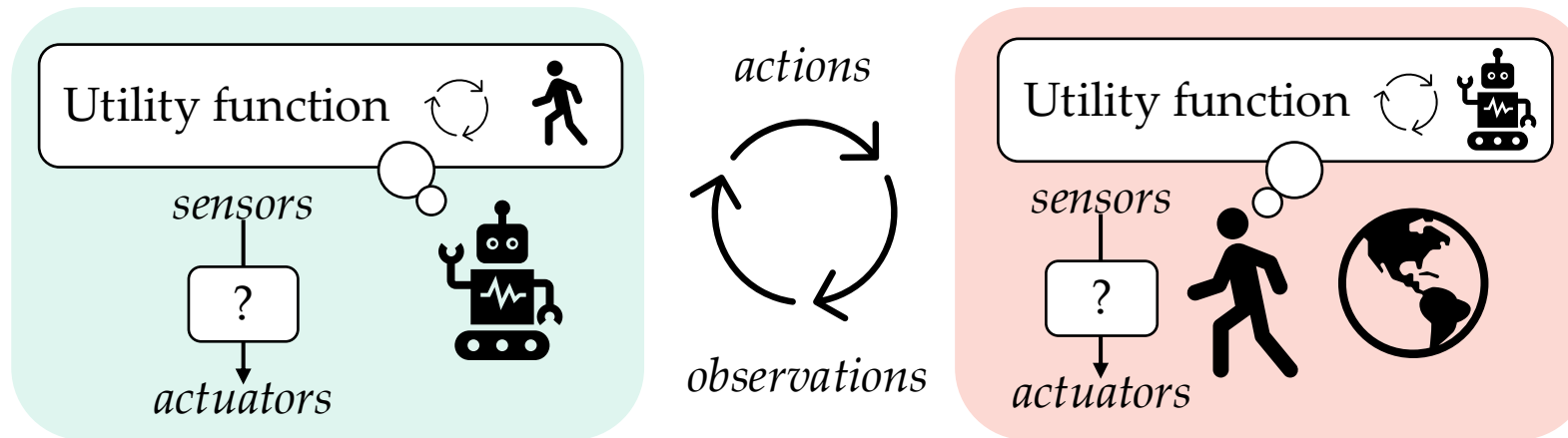
	See	Think	Act
Robotics	Detect objects, colors, estimate distances, poses, geometry, etc.		
HRI	Detect objects, colors, distances, poses, geometry, etc. Estimate human's utility, intent, preferences, strategy/style		

How is algorithmic HRI different?



	See	Think	Act
Robotics	Detect objects, colors, estimate distances, poses, geometry, etc.	Plan / optimize for own utility function	
HRI	Detect objects, colors, distances, poses, geometry, etc. Estimate human's utility, intent, preferences, strategy/style	Plan / optimize for • Own utility function • Adapting / coordinating with human • Influencing human belief about the world or robot	

How is algorithmic HRI different?



	See	Think	Act
Robotics	Detect objects, colors, estimate distances, poses, geometry, etc.	Plan / optimize for own utility function	Execute motor commands (e.g., based on physics models)
HRI	Detect objects, colors, distances, poses, geometry, etc. Estimate human's utility, intent, preferences, strategy/style	Plan / optimize for • Own utility function • Adapting / coordinating with human • Influencing human belief about the world or robot	May adjust execution dynamically based on human feedback (e.g., force feedback corrections)

But this seems really hard to encode into
our algorithms...



Where are people interacting with advanced autonomy the *most* right now?

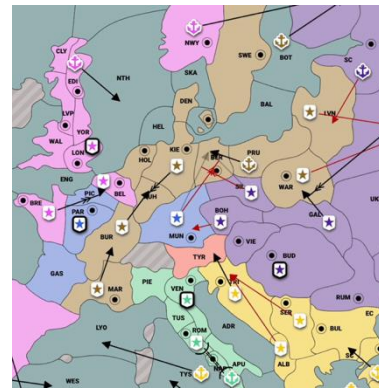


Exciting time for *interactive* Artificial Intelligence!

AlphaGo



CICERO



RESEARCH ARTICLE | COMPUTER SCIENCE
Human-level play in the game of *Diplomacy* by combining language models with strategic reasoning

META FUNDAMENTAL AI RESEARCH DIPLOMACY TEAM (FAIR) ANTON BAKHTIN · NOAM BROWN · EMILY DINAN · GABRIELE FARINA · COLIN FLAHERTY · DANIEL FRIED · ANDREW GOPE · JONATHAN GRAY · I-I AND MARKUS ZILSTRAS · +17 authors [Authors Info & Affiliations](#)

SCIENCE · 22 Nov 2023 · Vol 378, Issue 6624 · pp. 1067-1074 · DOI:10.1126/science.adr9097

105,222

CHECK ACCESS

AI masters Diplomacy

The game *Diplomacy* has been a major challenge for artificial intelligence (AI). Unlike other competitive games that AI has recently mastered, such as chess, Go, and poker *Diplomacy* cannot be solved purely through self-play: it requires the de-

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ChatGPT

2016

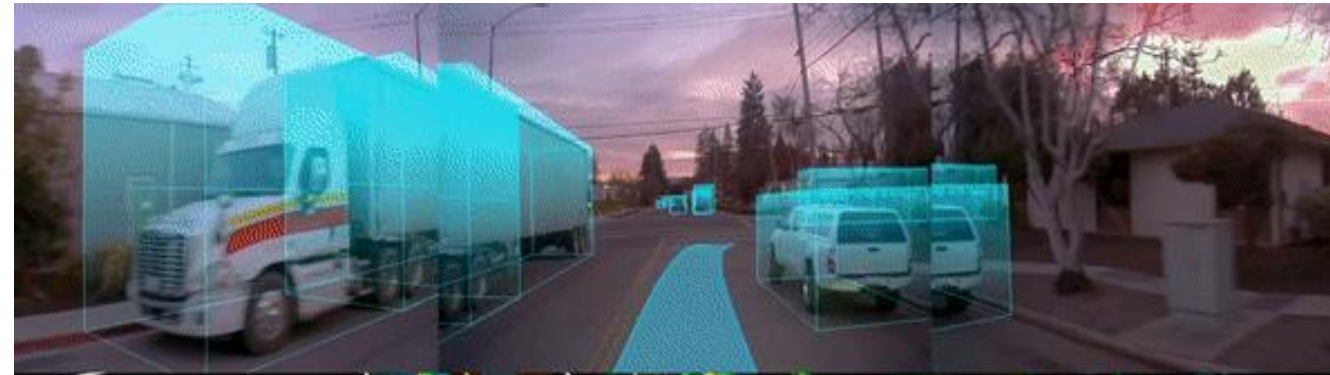
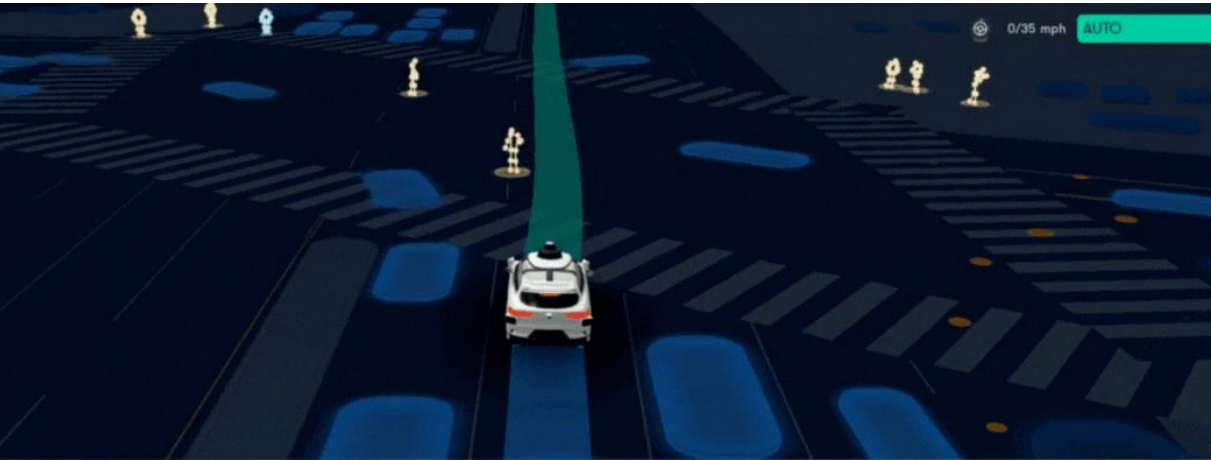
2022

today

But where are the interactive **robots**?



How do robots interact with people today?



How do robots interact with people today?



How do robots interact with people today?



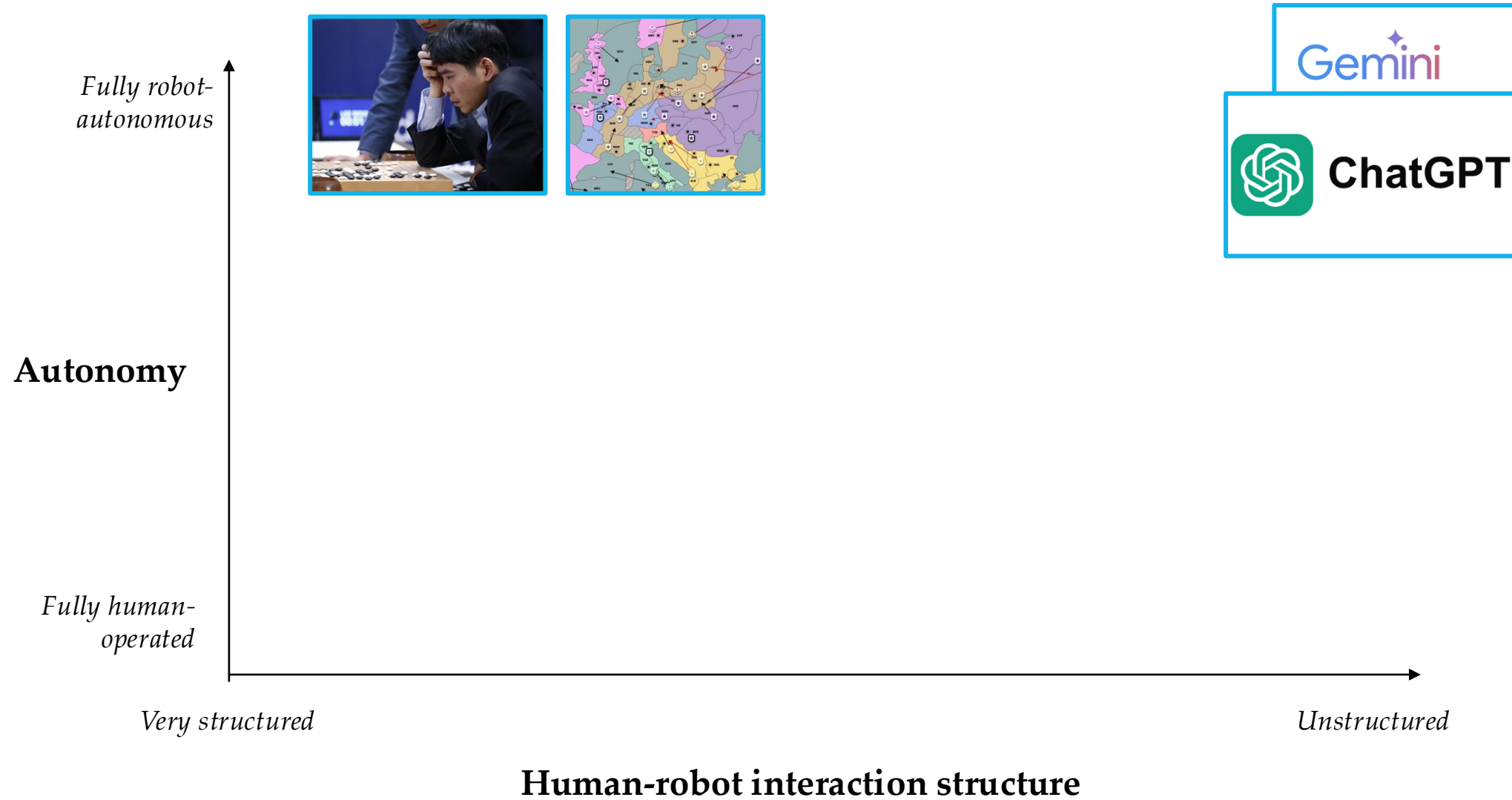
How do robots interact with people today?



hello robot™

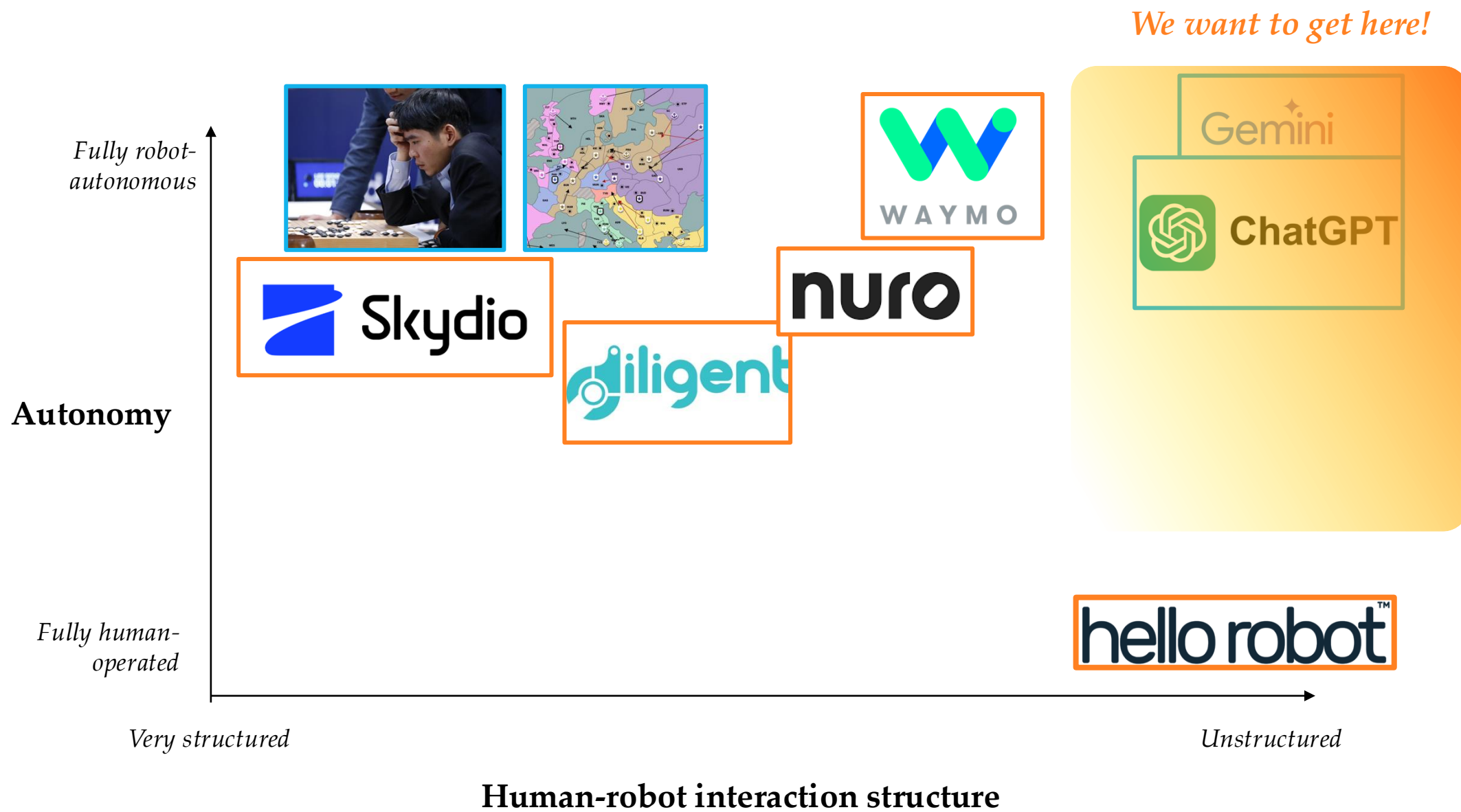


Here is where we are in AI...

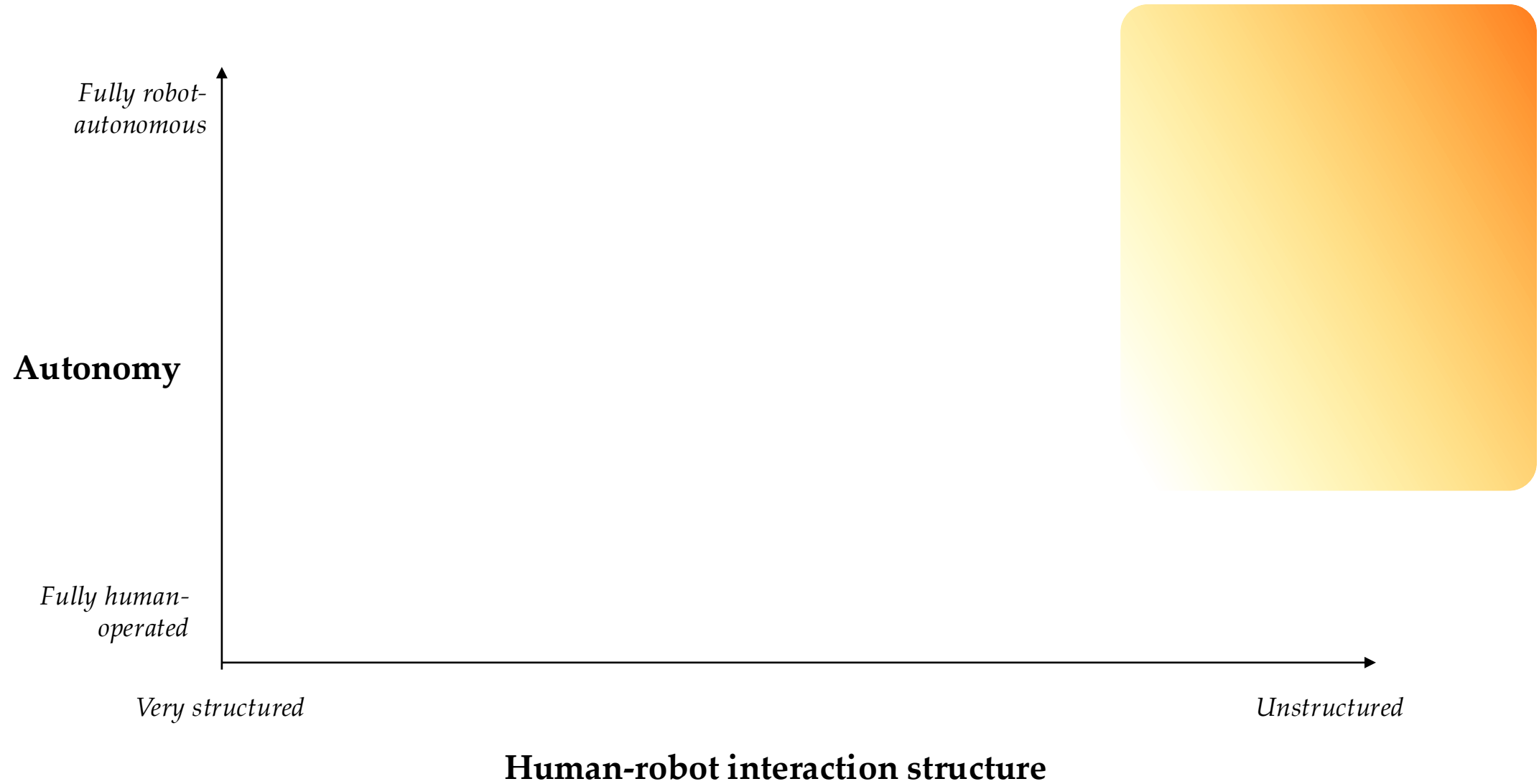


Here is where we are in robotics...



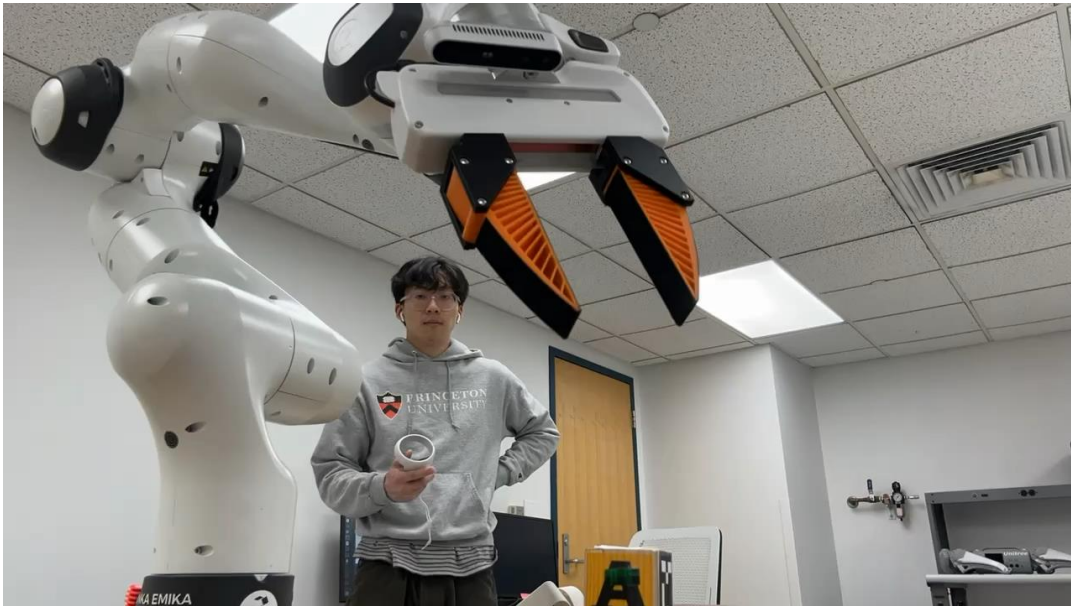


Think: Why are robots *not* in millions of homes?



1. The way we program robots is rigid

Not flexible enough to be used by everyday users for everyday tasks; requires expert knowledge



Engineers Design Behaviors

Ship robot



Users can't easily
expand capabilities,
or experience
unexpected failures!

2. Hard to specify to robots what matters to people

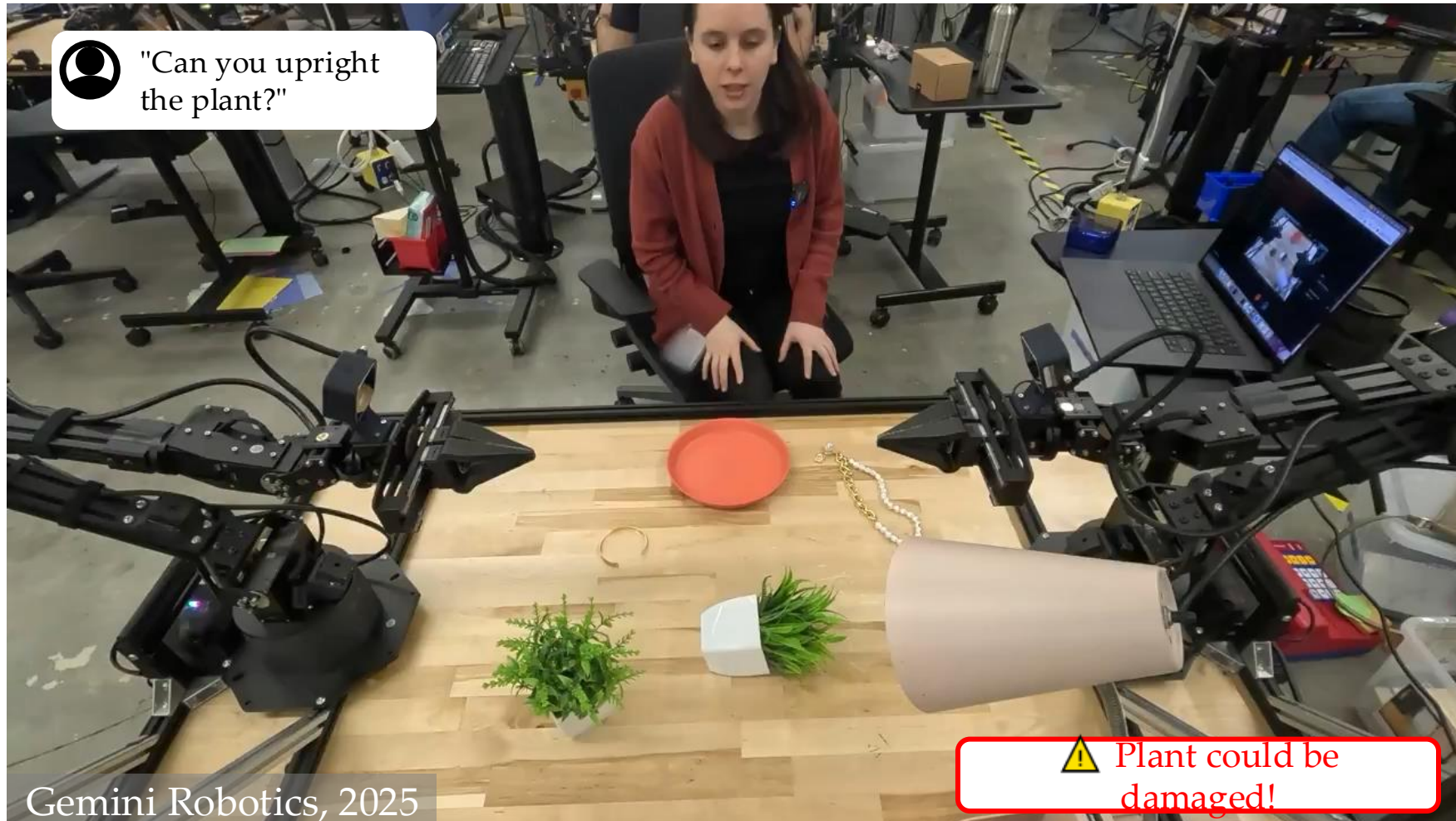
- *Autonomy*: hard to design robot policies that behave according to what end-users want
- *Evaluation*: hard to write metrics that correlate with what end-users want



“Feel the Bite: Robot-Assisted Inside-Mouth Bite Transfer using Robust Mouth Perception and Physical Interaction-Aware Control.” Jenamani, et al. (2024)

2. Hard to specify to robots what matters to people

This is also true in the “generalist” robot paradigm, not just assistive robotics!



3. Hard to model human interaction

Human behavior is diverse: varying between individuals, environments, and over time → long tail!



Why this course?

Take any robot application and ...



1) **Model / quantify** human interaction with robots

2) **Solve** robot decision-making algorithms that are informed of/by people

3) **Identify** the frontiers of human-robot interaction

What you will learn in this course

Foundations

- Single & multi-agent decision-making (MDPs, POMDPs, RL)
- Intent inference & expression
- Reward & policy learning
- Experimental design

Prediction for Action

- Trajectory forecasting
- Collaboration, assistance, shared autonomy
- Game theory

Learning, Alignment, and Safety

- World modeling
- Alignment
- Active learning
- HRI in era of foundation models
- Safety & uncertainty quantification

Guest Lectures

Trajectory Forecasting



Ingrid Navarro
Final-Year PhD Student @ CMU

Active Robot Learning



Erdem Biyik
Prof @ USC

HRI in the Era of Foundation Models



Guy Rosman
Toyota Research Institute



Jonathan DeCastro
Toyota Research Institute

General Resources

- No textbook!
- If I were to recommend textbooks for this class...

Artificial Intelligence: A Modern Approach by Russell and Norvig

Reinforcement Learning by Sutton and Barto

Probabilistic Robotics by Thrun, Burgard, Fox

Dynamic Noncooperative Game Theory by Başar and Olsder

Humans and Automation by Sheridan

Course Logistics

Format: lecture or related paper reading discussions

Typical 80-min class:

- 5-10 min logistics and recap

- 70 min lecture, invited talk, or paper discussion

Use *course website* for up-to-date schedule & paper links

<https://abajcsy.github.io/human-robot-interaction/>

Human Robot Interaction

Fall 2026. 16-867. Tuesday / Thursday 11:00am-12:20pm.



Announcements

Hello!

Jul 1 · 0 min read

See you next semester! 🤖

Course Overview

Schedule (Tentative)

Foundations

Aug. 25:	Course Overview	Syllabus
Aug. 27:	Single-Agent Decision-Making	
Sept. 1:	Value Iteration, Reinforcement Learning	Test Your Knowledge (from CS188, Berkeley)
Sept. 3:	POMDPs Project Ideation Exercise	
Sept. 8:	Probability, Bayesian Inference	Goal Inference as Inverse Planning
Sept. 10:	Intent Inference & Expression	HW #1 DUE Expressing Thought, Functional Expressive Motion, Predictability & Legibility
Sept. 15:	Reward and Policy Learning	An Invitation to Imitation , MaxEntIRL , SAILOR
Sept. 17:	Experimental Design & Statistical Analysis	PROJECT PROPOSAL DUE A Primer for Conducting Experiments in Human-Robot Interaction

Prediction for Action

Sept. 22:	Collaboration, Assistance, & Coordination	
Sept. 24:	GUEST LECTURE Trajectory Forecasting (Ingrid Navarro, CMU)	

Please read the *Syllabus* on the *course website*

<https://abajcsy.github.io/human-robot-interaction/>

Syllabus

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Overview

Robot interaction with people is inevitable: human engineers iteratively tune robot policies, autonomous cars navigate through crowded cities, construction workers teleoperate drones for building inspections, and assistive robots help end-users with daily living tasks. In this class we will formalize such human-robot interaction problems algorithmically. We will build the mathematical foundations for modeling human-robot interaction across robots and tasks, enable robots to understand human intent and predict human behavior, and study how robot learning changes in the presence of human feedback. The approaches covered will draw upon a variety of disciplines and tools such as sequential

Also: Helpful answers to *FAQs*!

<https://abajcsy.github.io/human-robot-interaction/>

FAQ

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- 1 [I don't have access to robots! What can I do?](#)
- 2 [I don't have access to compute! What can I do?](#)
- 3 [For my class project, I want to test an algorithm with people. Do I need to run a formal user study?](#)
- 4 [How can you effectively read a research paper?](#)
- 5 [How do you write a good research paper?](#)
- 6 [How do I make nice figures for a paper or talk?](#)

I don't have access to robots! What can I do?

If you want to use **physical hardware**, consider using the [AI Maker Space](#) in Tepper for your projects! To get more info, you can contact the AI Maker Space manager: Greg Armstrong at ai-makerspace@cs.cmu.edu.

If using physical hardware is infeasible for your project, consider using **simulation** or **benchmark datasets** for controlled testing and algorithm design. Some relevant resources include:

Autonomous Driving

- [CARLA](#) - driving simulator with urban layouts, buildings, vehicles, and sensors
- [CARLO](#) - a lightweight 2D version of CARLA

Use *Canvas* for downloading / uploading assignments

≡ 16867-A

Fall 2025

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Human Robot Interaction

 Assign To

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⋮

Welcome to **16-867: Human-Robot Interaction!**



Robot interaction with people is inevitable: human engineers iteratively tune robot policies, autonomous cars navigate through crowded cities, construction workers teleoperate drones for building inspections, and assistive robots help end-users with daily living tasks.

In this graduate class, we will formalize such human-robot interaction (HRI) problems algorithmically. We will build the mathematical foundations for modeling human-robot interaction across robots and tasks, enable robots to understand human intent and predict human behavior, and study how robot learning changes in the presence of human feedback. The approaches covered will draw upon a variety of disciplines and tools such as sequential decision-making, cognitive science, Bayesian inference, and modern machine learning. Throughout the class, there will also be several guest lectures from experts in the field. Students will practice essential research skills including reviewing papers, writing project proposals, and technical communication.

Grading

See class syllabus on course website for detailed info

Percentage	Activity
10%	Attendance & Participation
20%	HW (2x)
5%	Paper Summaries
5%	Project Proposal
20%	Midterm Project (Report + Presentation)
10%	Project Feedback from Course Staff
30%	Final Project (Report + Presentation)

Attendance & Participation (10%)

Expected to attend class in person—this is how we will all get the most out of the class! Please show up on time, especially for reading days

The way we grade this:

- First 5 minutes of class: we will give a **short, easy “quiz”** related to the last lecture’s content. This is graded as 1/0.
 - e.g., “*Describe what is a sequential decision-making problem.*”
- **Permitted 2 unexcused absences**, no questions asked, before being docked.

I understand that occasionally you may have challenges attending (e.g., illness, religious observance,..); **please let me know.**

Homework (20%)

HW #1: Foundations (written)

Released: *now!*
Due: Sept 10

HW #2: Reward Learning (coding)

Released: ~Nov. 2
Due: Nov. 15

These homeworks are *not* meant to be tedious; they are meant to **empower** you by showing you how to apply concepts from the class! 😊

If you are not confident (or are rusty) with Python and Pytorch, please come see us for educational resources!

Paper Summaries (5%)

Paper discussion days:

5 paper reading days

2 papers per reading day

Sept. 22:	Collaboration, Assistance, & Coordination	
Sept. 24:	GUEST LECTURE	Trajectory Forecasting (Ingrid Navarro, CMU)
Sept. 29:	Trajectory Forecasting	PAPER READING Confidence-Aware Prediction, ManiCast
Oct. 1:	HRI as a Game	Planning for AVs that Effect Humans, Contingency Games, iLQGames
Oct. 6:	Shared Autonomy	PAPER READING Policy Blending, LILA
Oct. 8:	Mid-term Project Presentations	
		MID-TERM PRESENTATION DUE

Before class:

Answer three questions about the paper:

- 1) What assumptions were made about (a) robot, (b) human, (c) their interaction
- 2) What extensions would you propose? Scope & justify
- 3) What do you like about this work?

Must submit on Canvas before class.

In class:

Split you into small groups, discuss set of questions, I assign a representative from each group to present on the group's takeaways, and the whole class can engage on the answer

Class Project

Two options:

Research project:

Identify a research direction broadly relevant to this class
Propose and take first steps towards an original idea

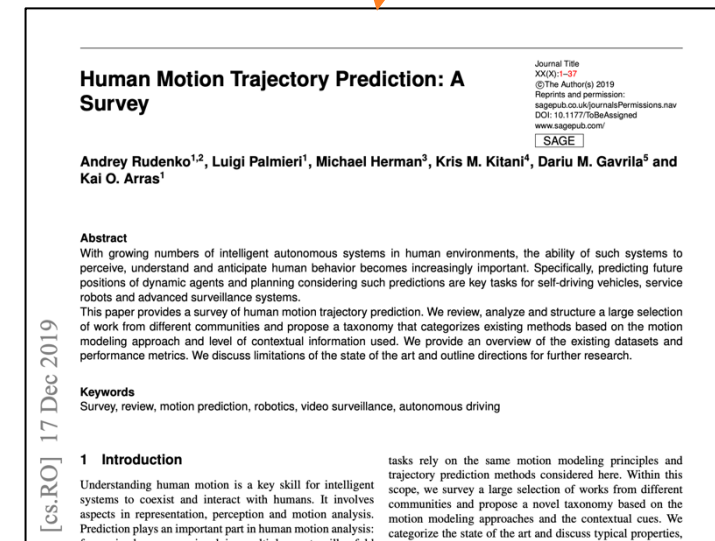
Literature survey:

Select a topic area and rigorous way in which you will find papers
Characterize this topic area in an insightful way (e.g., open questions, common assumptions, tractable vs. theoretical gaps)

Example of good literature survey



You must work in a group of min 2 to max 5 people. Explicit permission from Andrea must be granted if you want to work on an independent project



Class Project

When picking a project, make sure to answer the question:

How does the project connect to the broader topics & context of the class?

✓ Examples of projects within scope (non-exhaustive list!)

- Applying one of the techniques from class to your problem domain
 - *(e.g., using game-theory for pedestrian prediction, using RLHF to align an assistive AI agent, developing a new shared controller)*
- Rigorously comparing two methods that seek to solve the same problem
 - *(e.g., RLHF vs. DPO for aligning a diffusion policy, running a pilot human study to compare 2 shared controller designs)*
- Posing (and solving) a new algorithmic HRI problem for your problem domain
- Challenging an assumption underlying one of the methods in the class

Class Project

When picking a project, make sure to answer the question:

How does the project connect to the broader topics & context of the class?

When in doubt: come talk to us about your interests and we can help!

! Examples of projects not within scope

- No clear connection to human interaction
 - *e.g., a new self-supervised learning method for image classification; a robot motion planner for doing assembly tasks*
 - *To make them in scope: e.g., using human feedback to improve an image classification algorithm; robot motion planner which takes as input natural language descriptions of the assembly task*
- No clear connection to robotics / AI
 - *e.g., designing a new user interface for a mobile phone*
 - *To make them in scope: e.g., designing a new interface for humans to program robot behavior*

Class Project

Project Proposal (5%) – due: Sept. 17

1. **Report:** max 1 page project pitch

Mid-term Project (20%)

1. **Report:** max 4 page writeup -- due: Oct 22
2. **In-class Oral Presentation:** short conference-style project pitch (~3-5 mins) -- due: Oct 8

Final Project (30%)

1. **Report:** max 6 page writeup of findings
-- due: Dec 11
1. **In-class Oral Presentation:** short conference-style lightning talk pitch (~8-10 mins)
-- due: Nov 30

Class Project

Project Proposal (5%) – due: Sept. 17

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1. **Report:** max 6 page writeup of findings -- due: Dec 11
1. **In-class Oral Presentation:** short conference-style lightning talk pitch (~8-10 mins) -- due: Nov 30

Project Proposal

✓ Published

Assign To

Edit

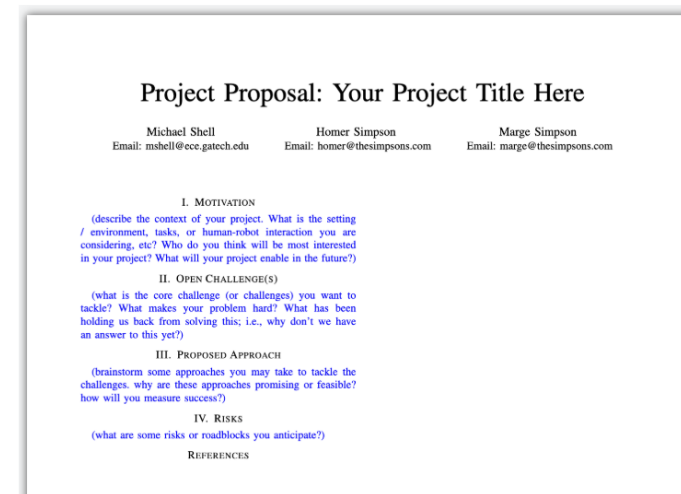
⋮

This is a brief (maximum 1 page, excluding references) project pitch. You can think of this as an extended abstract: you want to motivate the topic you have chosen and answer some key brainstorming questions about your directions.

Please use the attached Latex template and answer the questions below (which are also in the template):

- **Motivation:** describe the context of your project. What is the setting / environment, tasks, or human-robot interaction you are considering, etc? Who do you think will be most interested in your project? What will your project enable in the future?
- **Open Challenge(s):** what is the core challenge (or challenges) you want to tackle? What makes your problem hard? What has been holding us back from solving this; i.e., why don't we have an answer to this yet?
- **Proposed Approach:** brainstorm some approaches you may take to tackle the challenges. why are these approaches promising or feasible? how will you measure success?
- **Risks:** what are some risks or roadblocks you anticipate?

Latex Template (zip file): [project-proposal-latex.zip](#) ↓



Class Project

Project Proposal (5%) – due: Sept. 17

1. **Report:** max 1 page project pitch

Mid-term Project (20%)

1. **Report:** max 4 page writeup -- due: Oct 22 →
2. **In-class Oral Presentation:** short conference-style project pitch (~3-5 mins) -- due: Oct 8

Final Project (30%)

1. **Report:** max 6 page writeup of findings -- due: Dec 11
1. **In-class Oral Presentation:** short conference-style lightning talk pitch (~8-10 mins) -- due: Nov 30

Mid-term Report

✓ Published

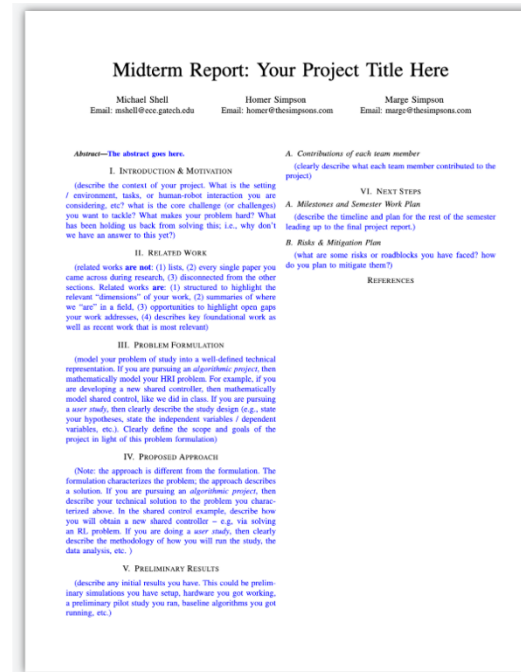
Assign To

Edit

⋮

This is intended as a checkpoint to ensure that you are making progress towards your final project. The report length should be a typical robotics workshop paper (maximum 4 pages, excluding references).

Please use the attached Latex template and follow the structure of the subsections.



Latex Template (zip file): [midterm-report-latex.zip](#) ↓

Class Project

Project Proposal (5%) – due: Sept. 17

1. **Report:** max 1 page project pitch

Mid-term Project (20%)

1. **Report:** max 4 page writeup -- due: Oct 22
2. **In-class Oral Presentation:** short conference-style project pitch (~3-5 mins) -- due: Oct 8

Final Project (30%)

1. **Report:** max 6 page writeup of findings —————→
-- due: Dec 11
1. **In-class Oral Presentation:** short conference-style lightning talk pitch (~8-10 mins)
-- due: Nov 30

Final Project Report

✓ Published

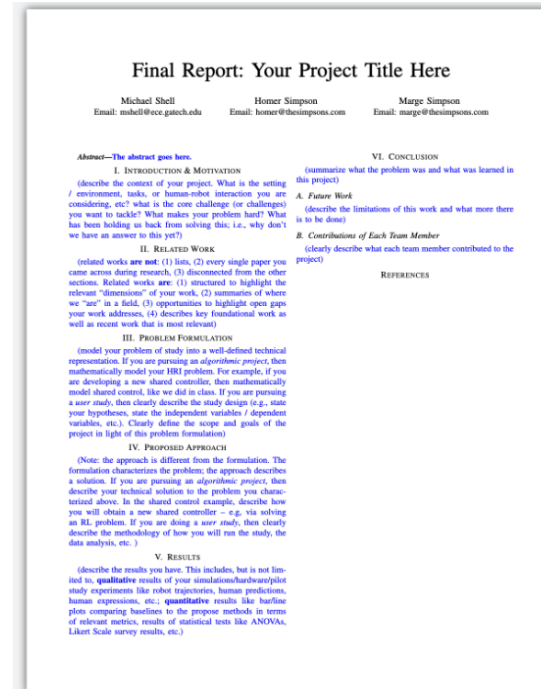
Assign To

Edit

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The final report should present your final findings in a research or survey paper format. The length should be maximum 6 pages, double-column. The grade will be determined based on the content quality and not on the absolute length (please see the grading rubric below).

Please use the attached Latex template and follow the structure of the subsections.



Latex Template (zip file): [final-report-latex.zip](#) ↓

Project Feedback from Course Staff (10%)

A final, but important part of the class is engaging with the course staff about your project throughout the semester.

You must schedule a 30 min meeting with a member of course staff sometime between the start of the semester and Nov. 20th (Friday) to receive detailed feedback.

After the meeting, please submit a brief summary of the discussion and submit it under the Canvas assignment for your grade.

Project Feedback from Course Staff

✓ Published

Assign To

Edit

A final, but important part of the class is engaging with the course staff about your project. *You must schedule a 30 min meeting with a member of course staff sometime between the start of the semester and Nov. 20th (Friday) to receive detailed feedback.* **After the meeting, please submit a brief summary of the discussion and submit it under this Canvas assignment for your grade.**

Please have all team members individually submit copies of the discussion summary, but identify who the team members are in the write-up.

Points100

Submittinga text entry box

<https://forms.gle/LFvwdLoy82UxGjjeA>

Survey (5 min)



16-867

Human Robot Interaction

Introduction

Instructor: Andrea Bajcsy